

Building a Recommender System From Scratch

11 methods on MovieLens, measured beyond accuracy.



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11
methods, one
interface

12
metrics tracked

Dataset & EDA

MovieLens Latest Small — a per-user **chronological** **80/20 split**, not random, so the model never trains on a user's future ratings.

98.3%

of the user-item matrix is empty

100,836

ratings · 610 users · 9,742 movies

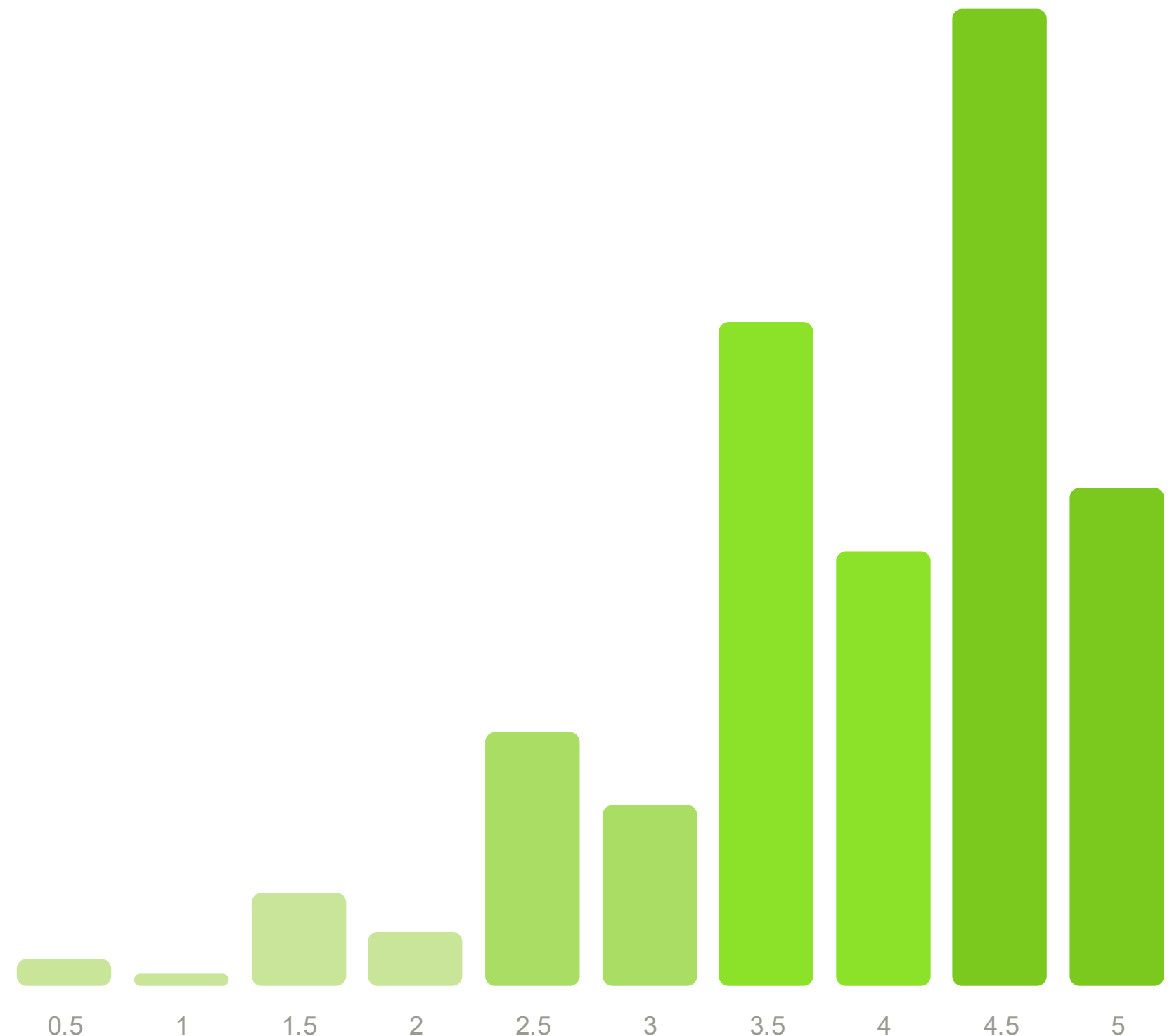
Generous raters. Ratings are left-skewed — most cluster at 3–4 stars.

A long tail. User activity and item popularity both follow a power law.

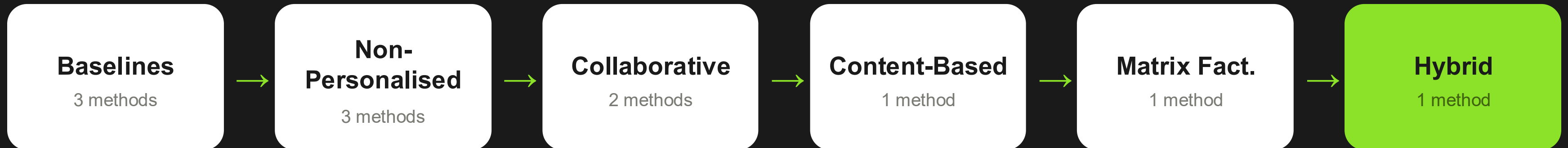
Genre concentration. Drama and Comedy dominate — the seed of popularity bias.

Rating distribution

count per star value · 0.5 → 5.0



One interface, eleven methods



A shared **fit / recommend / predict** contract (BaseRecommender ABC) runs the same data, metrics and test harness across every algorithm — so the comparison is genuinely fair.

0.769best MAE of all
11**0**Precision@1
0

The baseline paradox: User Average is the best **rating estimator** yet a useless **ranker**. MAE alone misleads.

Popularity, done carefully

$$WR(j) = v / (v+m) \cdot \bar{U}(j) + m / (v+m) \cdot C$$

Bayesian Weighted Rating

Shrinks small-sample items toward the global mean — 3 five-star ratings shouldn't outrank 500 ratings at 4.7.

1.000 → 0.255

popularity bias vs. raw Top-N

Product Associations

Lift = $P(X,Y) / [P(X) \cdot P(Y)]$ rewards genuinely surprising co-occurrences, not just common items.

0.011

lowest popularity bias of any method

Top-N Most Popular

Ranks by rating count. Competitive list quality — at the cost of maximum popularity bias.

1.000

every pick from the most-popular items

Collaborative filtering

User-Based CF

Pearson correlation over co-rated items only. Deviation-from-mean prediction normalises users who rate on different scales.

Why **k = 50** : precision peaks around 50 neighbours, then plateaus. More neighbours add noise, not signal.

Item-Based CF

Adjusted cosine: subtract each user's mean before the dot product, removing individual rating-scale bias so similarity reflects genuine preference.

Cached top-k item neighbourhood with a minimum-10-ratings filter.

0.012 > 0.003

Item-CF beats User-CF on Precision@10 — item neighbourhoods are more stable because items accumulate ratings faster than users do.

Features, factors & a hybrid

Content-Based

TF-IDF over tags + genres + year; user profile = centred-rating weighted sum of item vectors. IDF down-weights noisy common tags.

0.108

coverage — highest

0.077

diversity — lowest

Matrix Factorisation

Truncated SVD, 50 latent factors. Captures abstract patterns — "cerebral thrillers" — with no explicit features. The accuracy winner.

0.078

Precision@10 — best

0.979

popularity bias

Hybrid

SVD's ranking power, with Content-Based pulling picks away from the popularity cluster. A live α slider in the prototype.

$$\alpha \cdot \text{SVD} + (1-\alpha) \cdot \text{CB}, \quad \alpha=0.6$$

Beyond accuracy

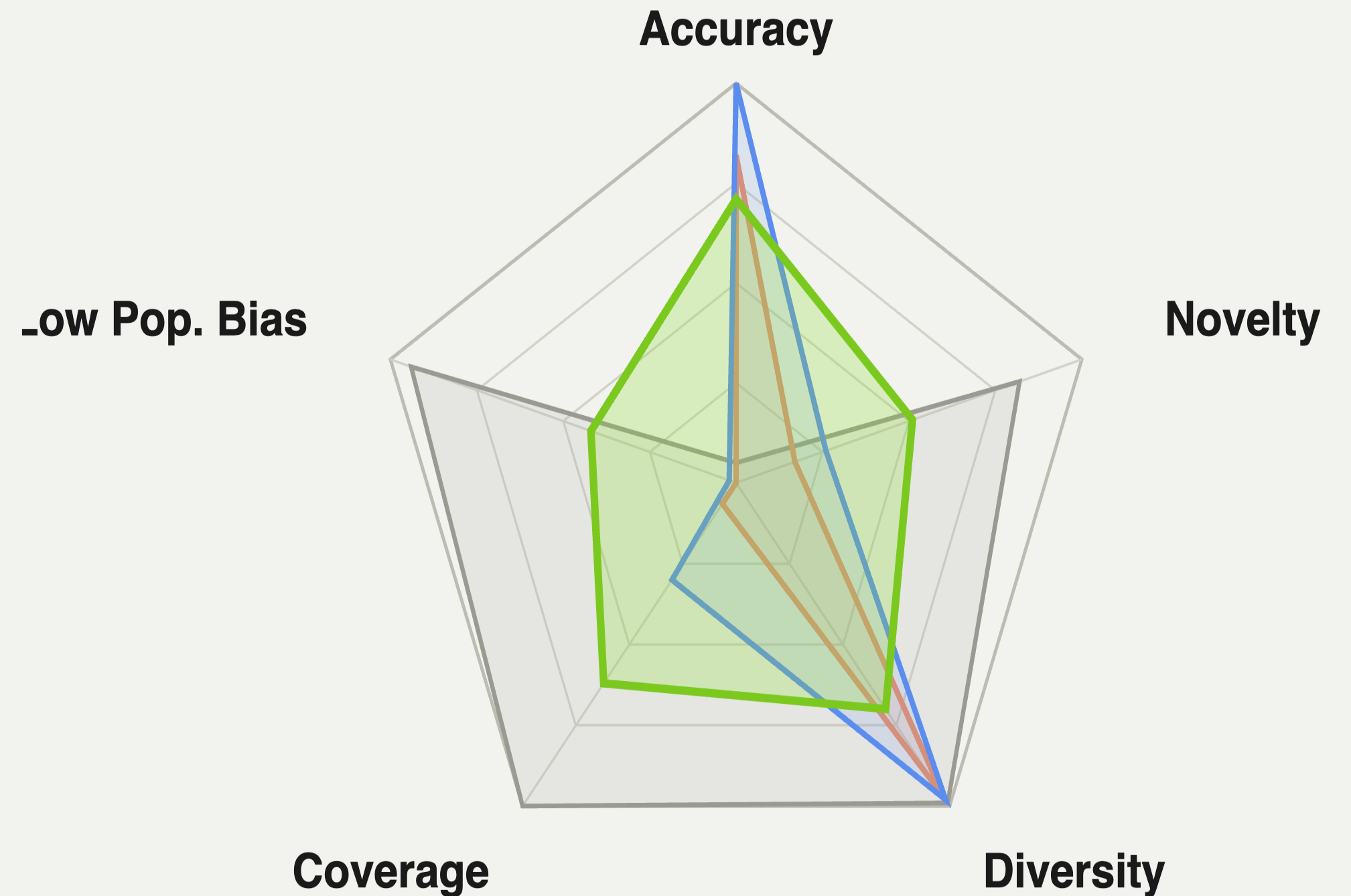
the accuracy leaders score worst on discovery · green = best in column

Method	Prec@10	NDCG	MAE	Coverage	Novelty	Pop.Bias	Serendip.
Random	0.002	0.005	1.617	0.115	7.62	0.062	0.002
User Average	0.000	0.000	0.769	0.001	9.25	0.000	0.000
Item Average	0.000	0.000	0.792	0.001	9.25	0.000	0.000
Top-N Most Popular	0.065	0.086	1.300	0.016	3.66	1.000	0.000
Bayesian Weighted	0.011	0.023	0.783	0.002	5.47	0.255	0.004
Product Associations	0.002	0.006	2.354	0.016	6.91	0.011	0.002
User-Based CF	0.003	0.004	0.767	0.051	6.34	0.130	0.003
Item-Based CF	0.012	0.014	0.756	0.057	4.58	0.436	0.012
Content-Based	0.002	0.003	1.243	0.108	7.68	0.072	0.002
Matrix Factorisation	0.078	0.105	0.761	0.034	2.44	0.979	0.053
Hybrid (SVD + CB)	0.043	0.075	0.873	0.071	4.68	0.582	0.030

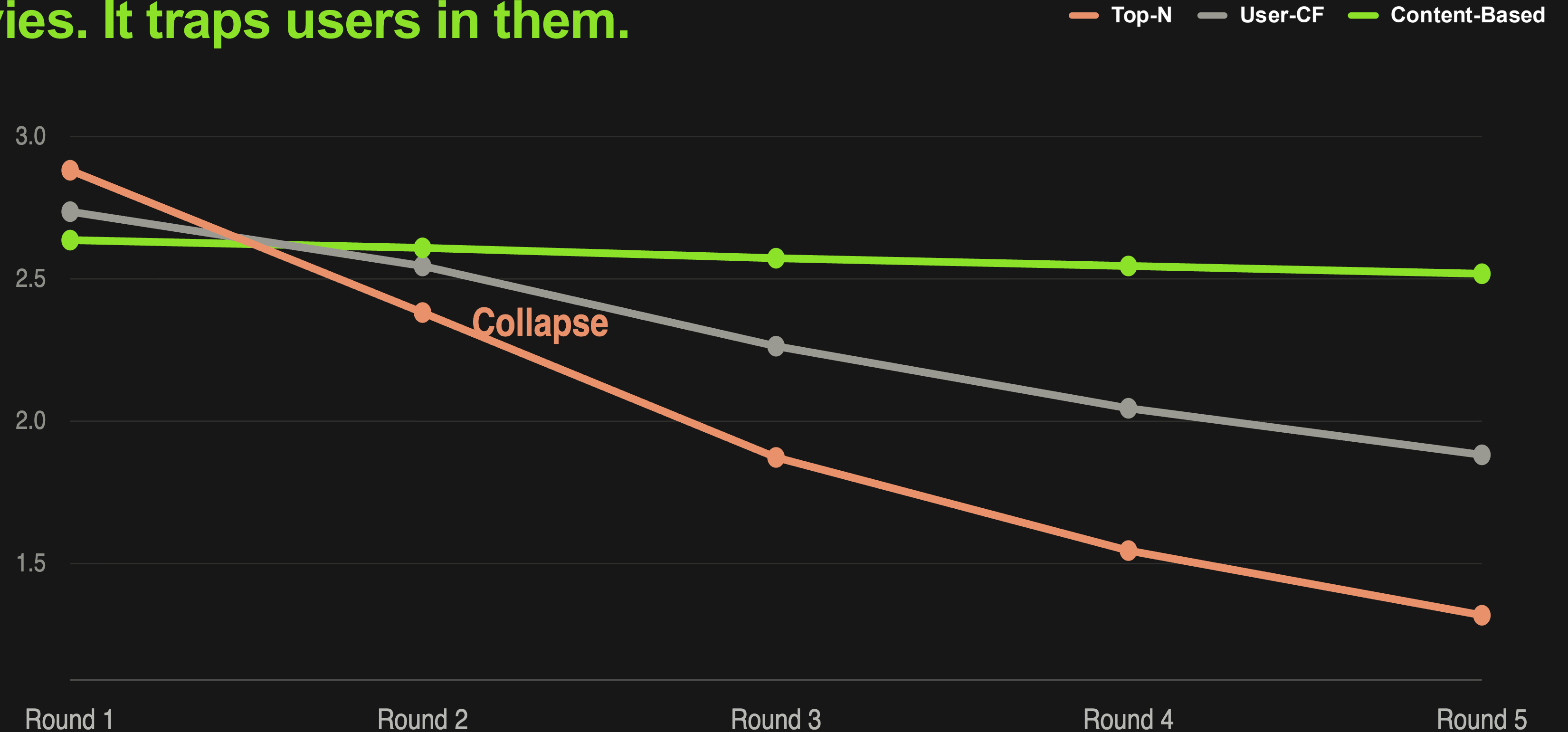
No method wins everything

Every method trades one strength for another. The radar makes it visible at a glance — and is the clearest justification for building all eleven.

- Hybrid — most balanced
- SVD — accuracy spike, discovery dip
- Top-N — popular, narrow
- Random — diverse, irrelevant



Top-N doesn't just recommend popular movies. It traps users in them.



Shannon genre entropy per simulated consumption round. Lower = narrower discovery.

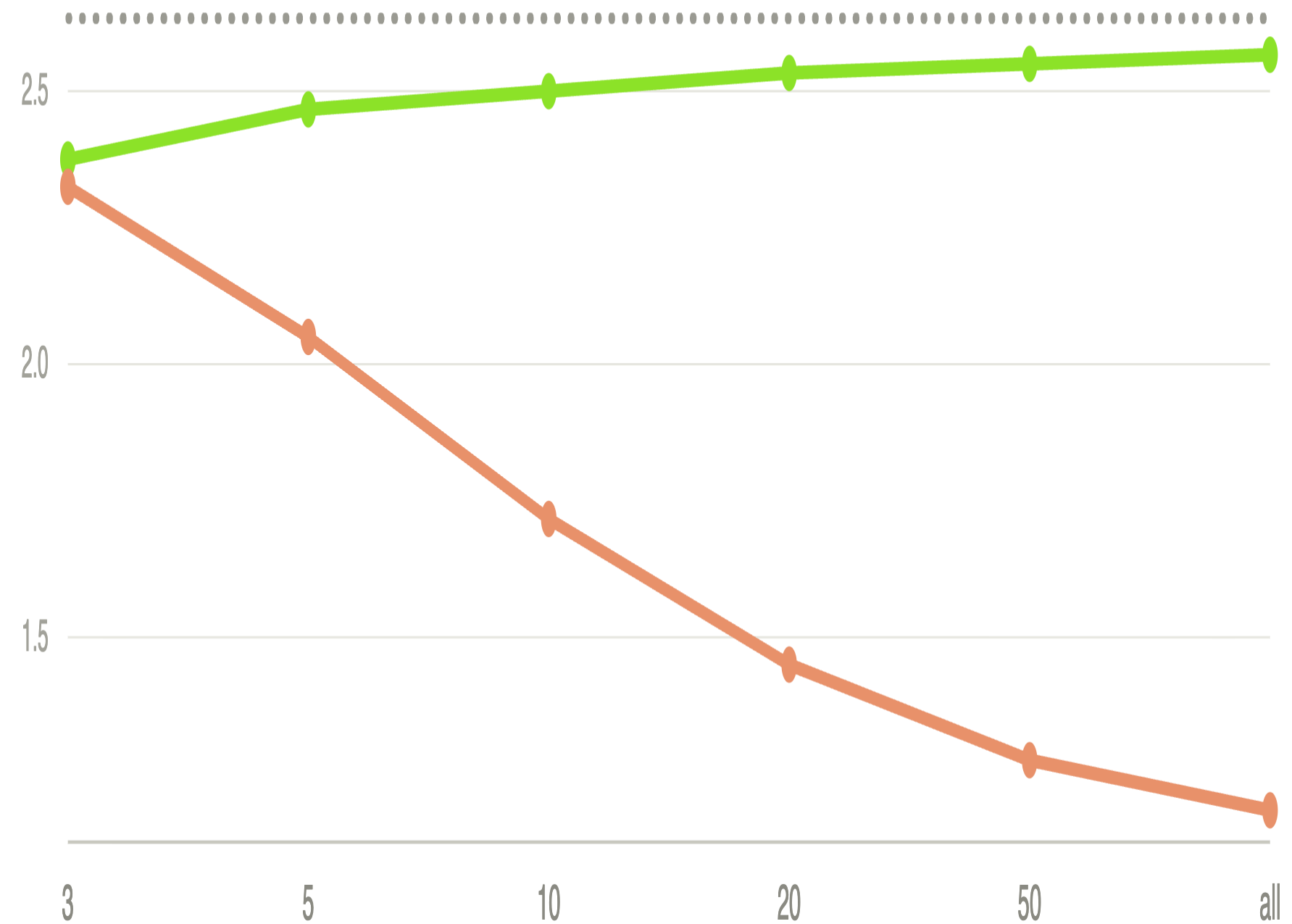
When does personalisation begin?

3

ratings — where Content-Based reaches reasonable diversity. It needs no neighbourhood.

The product play: personalise from signup with Content-Based, then hand off to the Hybrid as history grows. Cold start is a threshold, not a wall.

— Random — Content-Based — Item Average



Genre entropy of recommendations vs. ratings in user history.

What I built

A live α slider lets anyone drag between accuracy and discovery and watch the list change.

01

EDA

Grounds every claim — sparsity, the long tail, genre distribution.

02

Explorer

Recommendations + metrics side by side, with a live comparison radar.

03

Filter Bubble

The echo-chamber simulation — interactive and user-selectable.

04

By Genre

Netflix-style rows that make picks legible to anyone.

05

Cold Start

The personalisation-threshold experiment, at every history size.

Accuracy and discovery are in tension.

Optimise purely for precision and the system converges on the same popular items for everyone. Managing that tension — not eliminating it — is the real design challenge.

Production recommendation

Hybrid · $\alpha = 0.6$

0.043

Precision@10

0.030

Serendipity

0.071

Coverage

Two honest limitations

Serendipity is relative

Defined against Top-N — change the baseline and the number changes.

Offline $\neq$ enjoyment

Precision predicts held-out ratings, not whether a user returns.

If I took this further

- **Implicit feedback.** Move from explicit stars to watch time and skips — replacing SVD with ALS or BPR.
- **Explanations at scale.** "Because you watched X" — transparency as both ethics and product.
- **A/B test the diversity floor.** Does it improve long-term engagement, or do users prefer the familiar?

**Thank
you.**

